THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 162	Let breadth $= b$.
2	**(b)** $\frac{23}{24}$	Use the LCM of the denominators.
3	**(c)** touch externally	Compare 17 with 11+6=17 and |11-6|=5.
4	**(d)** 5	Number of sides $= 360 \div a$.
5	**(b)** 22	Simplify step by step.
6	**(c)** 16	Perimeters are equal.
7	**(b)** 60	Difference of the fractions, then divide.
8	**(a)** 20	Add the three sides.
9	**(b)** 45	Divide.
10	**(d)** 30	Opposites cancel.
11	**(d)** 120	Half of the rectangle.
12	**(b)** 40	Fraction removed = difference.
13	**(c)** 36	Side $= 2r$.
14	**(c)** 2	Letters H, I, O, X.
15	**(d)** 23	Distance $= \|a - b\|$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1).	1+1
17	Number $= 18 \div$ \frac{3}{4}$= 24$ (1); $\frac{1}{3}$ of the number $= 8$ (1).	1+1
18	Intersect when $4 < d < 16$ (1); Here d = 12: intersect at two points (1).	1+1
19	Sides $= 360 \div 30 = 12$ (1); Perimeter $= 12 \times 4 = 48$ cm (1).	1+1
20	Opposite integers cancel each other; Remaining: integers from 7 to 10 (1); Sum $= 34$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1).	1+1+1
22	Girls $=$ \frac{5}{8}$\times 480 = 300$ (1); Chess girls $=$ \frac{2}{5}$\times 300 = 120$ (1); Boys $= 480 - 300 = 180$ (1).	1+1+1
23	Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm^2 (1).	1+1+1
24	Angle $= 360 \div 5 = 72^\circ$ (1); Lines of symmetry $= 5$ (1); Total around centre $= 5 \times 72 = 360^\circ$ (1).	1+1+1
	OR: After dive: -80; After rise: -40 (1); After second dive: -65 (1); Total distance $= 115$ m (1).	1+1+1
25	Rectangle area $= 144$ cm^2; Square side $= \sqrt{144} = 12$ cm (1); Rectangle perimeter 52, square perimeter 48 (1); Square has the smaller perimeter (1).	1+1+1
	OR: $\frac{x}{12} =$ \frac{5}{6}$-$ \frac{1}{4} (1); $=\frac{7}{12}$ (1); $x = 7$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Rectangle area $= 240$ cm^2 (1); Triangle area $= \frac{1}{2} \times 20 \times 8 = 80$ cm^2 (1); Remaining $= 160$ cm^2 (1); Fraction left $= \frac{160}{240}$ (2).	1+1+1+2
	OR: After day 1: \frac{2}{3} left (1); Day 2 sells $\frac{1}{6}$ of total (1); Left fraction $\frac{1}{2}$ (1); Total $= 30 \div \frac{1}{2} = 60$ m (2).	1+1+1+2
27	Draw AB = 6 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 8 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(6 + 8) = 28$ cm (1).	1+1+1+1+1
	OR: (a) 6 (1); (b) 60° (1); (c) 30 cm (1); (d) 240° (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 336	Outer minus inner.
2	**(b)** 8	Divide the length by the piece length.
3	**(d)** 16	Two diameters.
4	**(d)** 150	Add the smallest angles.
5	**(a)** -17	Subtract.
6	**(d)** 14	Perimeter does not change.
7	**(c)** 5	Find the number first.
8	**(a)** 16	Each is half the diagonal.
9	**(d)** 3	If a figure has a line and order n, it has n lines.
10	**(c)** -30	Gains positive, losses negative.
11	**(d)** 45	Area = cost ÷ rate.
12	**(c)** $\frac{3}{4}$	Compare with a common denominator.
13	**(b)** 18	Each side is $2r$.
14	**(a)** -3	Change the sign of x.
15	**(c)** -6	The first two cancel.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area = 1200 m^2, perimeter = 140 m; Levelling cost = 6000 (1); Fencing cost = 1680; total = 7680 (1).	1+1
17	Required =\frac{5}{6}−\frac{2}{7} (1); = $\frac{23}{42}$ (1).	1+1
18	$\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1).	1+1
19	Order = $360 \div 40 = 9$ (1); 120° is a multiple: 3 times 40 (1).	1+1
20	$x = 5 - (-9)$ (1); $x = 14$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Longest chord = diameter = $2 \times 9 = 18$ cm (1); Width of ring = 5 cm (1); Inner diameter = 8 cm (1).	1+1+1
22	A' has x = $2 \times 5 - 1 = 9$, y = 2 (1); B' has x = $2 \times 5 - 4 = 6$, y = 2 (1); y-coordinates do not change (1).	1+1+1
23	Other = $-4 - 11 = -15$ (1); Difference = 26 (1); Check the sum: -4 (1).	1+1+1
24	Floor in cm: 600 by 400; Tile area 600 cm^2 (1); Floor area 240000 cm^2 (1); Tiles = 400 (1).	1+1+1
	OR: Together =\frac{1}{4}+\frac{1}{6}= $\frac{5}{12}$ (1); Remaining = $1 - \frac{5}{12} = \frac{7}{12}$ (1); State the answer (1).	1+1+1
25	BM = 5 cm, BN = 3 cm (1); Perimeter = $2(5 + 3) = 16$ cm (1); Area = 15 cm^2 (1).	1+1+1
	OR: Smallest angle = $360 \div 6 = 60^\circ$ (1); $180 \div 60 = 3$ rotations (1); A whole number means the figure fits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Scores: -3, 3, -6, 6 (1); Final score 6 (1); Lowest score -6 (1); Write all values as integers (2).	1+1+1+2
	OR: Path parallel to length: $60 \times 3 = 180$ m^2 (1); Path parallel to breadth: $40 \times 3 = 120$ m^2 (1); Common square = $3 \times 3 = 9$ m^2 (1); Area of paths = 291 m^2 (1); Rest of the park = 2109 m^2 (1).	1+1+1+1+1
27	Sum of whole parts = 6 (1); Sum of fractions = 1 (1); Total = 7 (1); After subtracting: $6\frac{1}{4}$ (2).	1+1+1+2
	OR: Row length = $5 \times 6 = 30$ cm (1); First to last centres = $(5 - 1) \times 6 = 24$ cm (1); Circles in the strip = 7 (1); Show the working with diameter (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 64	A square has the greatest area for a given perimeter.
2	**(c)** $2\frac{3}{4}$	Subtract.
3	**(c)** 70	Base angles are equal.
4	**(b)** 9	$360 \div$ angle.
5	**(b)** -5	Add b to both sides.
6	**(b)** 42	Add the three sides.
7	**(d)** 24	LCM.
8	**(b)** 10	Diameter equals the side.
9	**(b)** 42	Number of sides times side.
10	**(b)** 2	Subtracting a negative adds.
11	**(a)** 84	Outer side $= a + 2$.
12	**(c)** $1\frac{1}{6}$	Use the LCM of the denominators.
13	**(b)** touch internally	Compare 8 with 13+5=18 and |13-5|=8.
14	**(c)** 10	Equal distance on the other side.
15	**(c)** 7	All pairs cancel except the last.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Room area $= 48$ m^2; Carpet area $= 20$ m^2 (1); Uncovered $= 28$ m^2 (1).	1+1
17	One way $= \frac{23}{20}$ km (1); Total $= 2\times = \frac{23}{10}$ km (1).	1+1
18	Area of square $= 100$ cm^2 (1); Area of AOB $= 25$ cm^2 (one quarter) (1).	1+1
19	(a) $(-2, 5)$ (1); (b) $(2, -5)$ (1).	1+1
20	Final: $-20 + 60 - 25 = 15$ m (1); Distance from 30 m: 15 m (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Common difference $= 6$ (1); Next terms: 4, 10, 16 (1); First positive term: 4 (1).	1+1+1
22	Perimeter $= 160$ m; Wire needed $= 2 \times 160 = 320$ m (1); Fencing cost $= 4800$ (1); Ploughing cost $= 7500$ (1).	1+1+1
23	Girls $=\frac{4}{9}\times 360 = 160$ (1); Chess girls $=\frac{3}{8}\times 160 = 60$ (1); Boys $= 360 - 160 = 200$ (1).	1+1+1
24	Distance between adjacent centres $= 2r = 8$ cm (1); Side $= 8$ cm (1); Area $= 64$ cm^2 (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); Lines of symmetry $= 6$ (1); Total around centre $= 6 \times 60 = 360^\circ$ (1).	1+1+1
25	After dive: -80; After rise: -25 (1); After second dive: -55 (1); Total distance $= 155$ m (1).	1+1+1
	OR: Rectangle area $= 324$ cm^2; Square side $= \sqrt{324} = 18$ cm (1); Rectangle perimeter 78, square perimeter 72 (1); Square has the smaller perimeter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Angles: 120° and 90° (1); Difference 30° (1); LCM of 120 and 90 is 360 (1); Common turn 360° (2).	1+1+1+2
	OR: After down: -2 (1); After up: 0 (1); After down: -3 (1); Exists (2).	1+1+1+2
27	Rectangle area $= 240$ cm^2 (1); Triangle area $= \frac{1}{2} \times 24 \times 6 = 72$ cm^2 (1); Remaining $= 168$ cm^2 (1); Fraction left $= \frac{168}{240}$ (2).	1+1+1+2
	OR: After day 1: \frac{3}{4} left (1); Day 2 sells $\frac{1}{2}$ of total (1); Left fraction $\frac{1}{4}$ (1); Total $= 12 \div \frac{1}{4} = 48$ m (2).	1+1+1+2

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